What is the difference between "trivial" and "non-trivial" solutions of a linear system?

Answer:

The distinction between "trivial" and "non-trivial" solutions is used almost exclusively for a specific type of system: a **homogeneous linear system**.

A **homogeneous system** is one where all the constant terms are zero. It can be written in the form Ax = 0, where A is a matrix of coefficients, x is a vector of variables, and 0 is the zero vector.

Here's the breakdown of the difference:

The Trivial Solution

The **trivial solution** to a homogeneous linear system is the solution where all variables are equal to zero. $x_1 = 0, x_2 = 0, \dots, x_n = 0$. This solution is called "trivial" because it is always a solution to any homogeneous system, and it's obvious without doing any work. If you substitute zeros for all the variables, each equation will simply become 0 = 0, which is always true.

Example: For the homogeneous system: $\begin{cases} x + 5y - z & = 0 \\ 2x - y + 3z & = 0 \end{cases}$

The trivial solution is (x, y, z) = (0, 0, 0). Plugging it in:

1. (0) + 5(0) - (0) = 0 (True)
2. 2(0) - (0) + 3(0) = 0 (True)

Because the trivial solution always exists for homogeneous systems, the real question is not "Is there a solution?" but rather, "Is the trivial solution the *only* solution?" This leads us to non-trivial solutions.

A **non-trivial solution** is any solution to a homogeneous linear system where at least one variable is non-zero.

The existence of non-trivial solutions is far more interesting. It tells us something important about the relationships between the equations (and the vectors or columns in the matrix). A homogeneous system will have non-trivial solutions if and only if there is at least one variable in the system that correspond to a non-pivot column after you reduce it to echelon form.

An Example

$$\begin{cases} x + y - 2z & = 0 \\ 2x + 3y - 5z & = 0 \end{cases}$$

1. Augmented Matrix: $\begin{bmatrix} 1 & 1 & -2 & 0 \\ 2 & 3 & -5 & 0 \end{bmatrix}$

2. Reduce to RREF: $\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \end{bmatrix}$

Column 3 is non-pivot, so the variable z can be assigned as parameter z = t. This means the homogeneous system has non-trivial solutions.

3. Analyze the Result: The RREF gives us the equations:

1. x - z = 0
2. y - z = 0

To find the General Solution:

- From eq (1): x - t = 0 => x = t
- From eq (2): y - t = 0 => y = t

The general solution is (x, y, z) = (t, t, t).

- If we choose t = 0, we get the trivial solution: (0, 0, 0).
- If we choose any other value for t (e.g., t = 1), we get a non-trivial solution: (1, 1, 1). If t = -5, we get another non-trivial solution: (-5, -5, -5). In fact, there are infinitely many non-trivial solutions.

Summary of Conditions

Feature	**Trivial Solution**	**Non-Trivial Solutions**
Definition	The solution where all variables are zero.	Any solution where at least one variable is non-zero.
Existence	Always exists for any homogeneous system.	Only exist if the system has at least one variable that corresponds to non-pivot column of the REF.
Number of Solutions	If a homogeneous system has *only* one solution, then the unique solution must be the trivial solution.	If a homogeneous system has *more than* one solution, then it will have one trivial solution and infinitely many non-trivial solutions.
Condition on Matrix	When every column of the REF/RREF is a pivot column, except the augmented column, then the system has only the trivial solution.	When there is at least one non-pivot column in the REF/RREF, then the system has one trivial solution and infinitely many non-trivial solutions.

Note on Non-homogeneous Systems: The terminology of “trivial” and “non-trivial” solution is not used for non-homogeneous systems (where Ax = b and b ≠ 0). For those systems, the zero vector is not a solution, so there is no "trivial" case to begin with. We do not refer to the solutions of non-homogeneous system as non-trivial solutions.

Scenario 1

Given the homogeneous System with 3 Equations and 2 Variables $\begin{cases} x + 2y & = 0 \\ 3x - y & = 0 \\ x + 7y & = 0 \end{cases}$

Augmented Matrix: $\begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 0 \\ 1 & 7 & 0 \end{bmatrix}$

After performing Gauss-Jordan elimination, you arrive at the RREF: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

Interpretation: The two columns corresponding to variables x and y are both pivot columns. This shows that the homogeneous system has only the trivial solution. (x = 0, y = 0).

Scenario 2

Given the Homogeneous System with 3 Equations and 3 Variables $\begin{cases} x + y + z &= 0 \\ x + 2y + 3z &= 0 \\ 2x + 3y + 4z &= 0 \end{cases}$

Augmented Matrix: $\begin{bmatrix} 1 & 1 & 1 & 0 \\ 1 & 2 & 3 & 0 \\ 2 & 3 & 4 & 0 \end{bmatrix}$

After performing Gauss-Jordan elimination, you arrive at the RREF $\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

Interpretation: The column for variable z is a non-pivot column, making z a free parameter. This is the condition that guarantees infinitely many non-trivial solutions in addition to the trivial solution.

Scenario 3

Given a non-Homogeneous System with 3 Equations and 3 Variables $\begin{cases} x + y + z &= 6 \\ x - y + 2z &= 5 \\ 2x + y - z &= 1 \end{cases}$

Augmented Matrix: $\begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & -1 & 2 & 5 \\ 2 & 1 & -1 & 1 \end{bmatrix}$

After performing Gauss-Jordan elimination, you arrive at the RREF $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{bmatrix}$

Interpretation: The left side is the identity matrix, which means all the three variables correspond to pivot columns. This system has a unique solution, which can be read directly from the RREF: x = 1, y = 2, and z = 3. However, this unique solution is not a "trivial" solution. In fact, since the system is non-homogeneous, it is irrelevant to refer to its solutions as "trivial" or "non-trivial".

Scenario 4

Given any homogeneous system with m equations and n variables where m < n. The augmented matrix is of size m x (n+1). Since there can be at most m pivot (leading entries) in the REF, it means there is at least one variable that corresponds to a non-pivot column of the REF. So the system must have infinitely many solutions, and hence will have non-trivial solutions (in addition to the trivial solution).